

Embedded System Design

UNIT 3

Anuj Kumar Sharma
Assistant Professor
Dept. of ECE
FET, GKV
Haridwar

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- Discussion of Syllabus of unit 3
- Third lab Experiment
- **Introduction to Microcontrollers and Micoprocessors,**
- Embedded versus external memory devices,
- CISC and RISC processors, General purpose processors.
- Harvard and Von Neumann Architectures.
- 8051 Microcontrollers- Assembly language.
- Architecture of 8051 Microcontroller.
- Registers , Addressing modes, Instruction set, I/O ports and memory organization.
- Interrupts Timer/counter and serial communication.

Third Lab Experiment

- **Object** : - To interface 16X2 LCD with Microcontroller 8051 and Implement it on Proteus and Breadboard.
- **Components required** : LCD , Microcontroller, resistors , capacitors ,Preset, Rom Burner , Kiel compiler , Proteus software , Connecting wires.
- **Theory** :- A 16x2 LCD means it can display 16 characters per line and there are 2 such lines. In this LCD each character is displayed in 5x7 pixel matrix. This LCD has two registers.

Conti....

- 1. Command/Instruction Register - stores the command instructions given to the LCD. A command is an instruction given to LCD to do a predefined task like initializing, clearing the screen, setting the cursor position, controlling display etc.
- 2. Data Register - stores the data to be displayed on the LCD. The data is the ASCII value of the character to be displayed on the LCD.

Conti...

- Commonly used LCD Command codes:



Hex Code	Command to LCD Instruction Register
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- | | |
|-------|---|
| ● 1 | Clear screen display |
| ● 2 | Return home |
| ● 4 | Decrement cursor |
| ● 6 | Increment cursor |
| ● E | Display ON, Cursor ON |
| ● 80` | Force the cursor to the beginning of the 1st line |
| ● C0 | Force cursor to the beginning of the 2nd line |
| ● 38 | Use 2 lines and 5x7 matrix |



Conti...

- The pin description of this module is given below.
- Pin configuration:
-
-
- Pin Symbol Description
- 1 VSS Ground 0 V
- 2 VCC Main power supply +5 V
- 3 VEE Power supply to control contrast Contrast adjustment by providing a variable resistor through VCC
- 4 RS Register Select
- RS=0 to select Command Register
- RS=1 to select Data Register
- 5 R/W Read/write
- R/W=0 to write to the register
- R/W=1 to read from the register
- 6 EN Enable A high to low pulse (minimum 450ns wide) is given when data is sent to data pins
- 7 DB0 To display letters or numbers, their ASCII codes are sent to data pins (with RS=1). Also instruction command codes are sent to these pins.
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Conti...

- 8 DB1
 - 9 DB2
 - 10 DB3
 - 11 DB4
 - 12 DB5
 - 13 DB6
 - 14 DB7
 - 15 Led+ Backlight VCC +5 V
 - 16 Led- Backlight Ground0 V
- 8-bit data pins

Conti...

- All the character lcd's of any size 8×1 , 8×2 , 10×2 , 16×1 , 16×2 , 16×4 , 20×2 , 20×4 , 24×2 , 30×2 , 32×2 , 40×2 , 40×4 contains HD44780 lcd controller in them. If not HD4478 then a controller compatible to it. HD44780 is responsible to accept data from external controller, verify data if correct then recognize if its a data to display on lcd screen or its a command to set lcd parameters. HD44780 has fixed and countable commands to initialize character lcd.

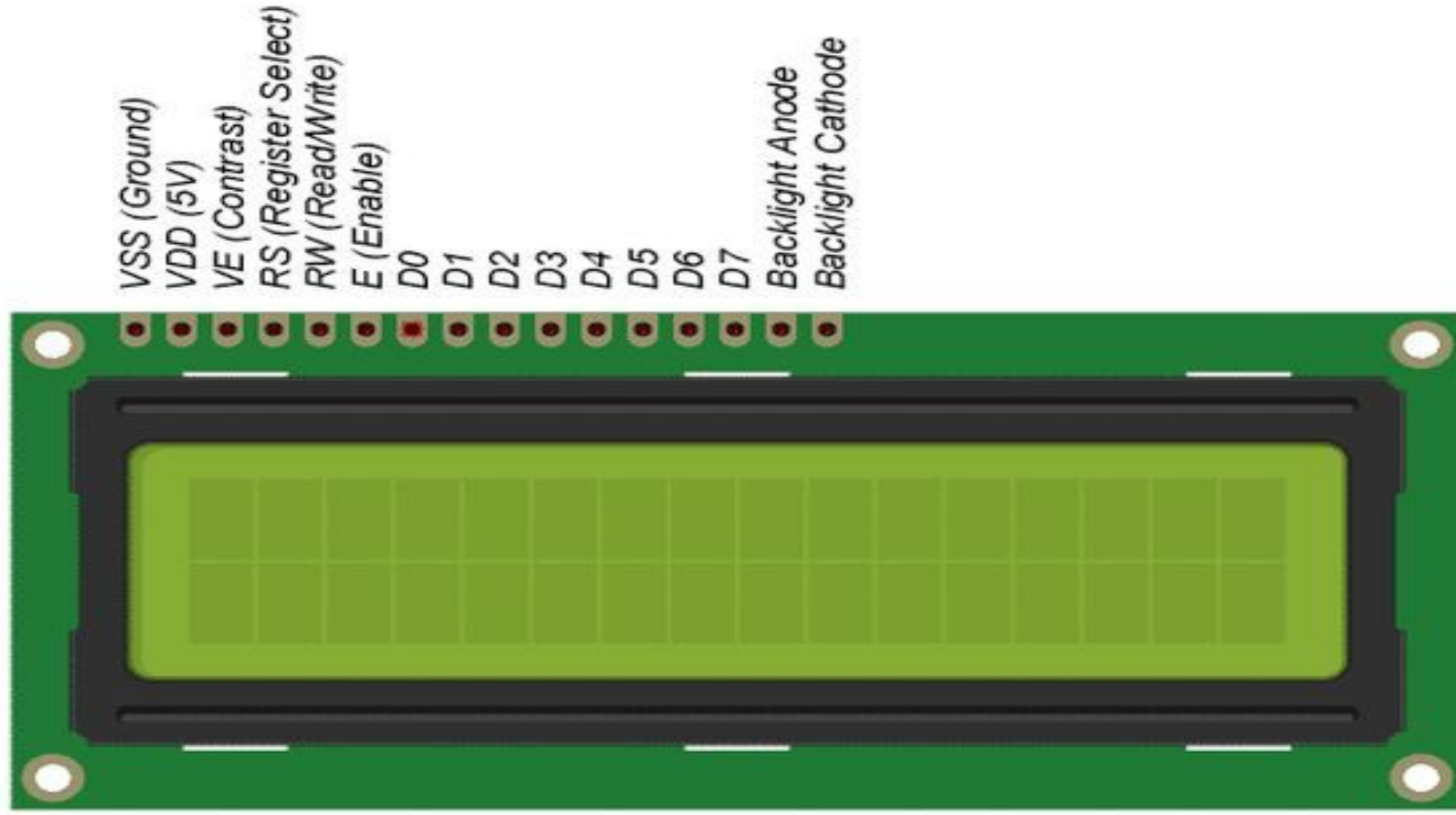
Programming the LCD:

- 1. Data pin8 (DB7) of the LCD is busy flag and is read when $R/W = 1$ & $RS = 0$. When busy flag=1, it means that LCD is not ready to accept data since it is busy with the internal operations. Therefore before passing any data to LCD, its command register should be read and busy flag should be checked.
- 2. To send data on the LCD, data is first written to the data pins with $R/W = 0$ (to specify the write operation) and $RS = 1$ (to select the data register). A high to low pulse is given at EN pin when data is sent. Each write operation is performed on the positive edge of the Enable signal.
- 3. To send a command on the LCD, a particular command is first specified to the data pins with $R/W = 0$ (to specify the write operation) and $RS = 0$ (to select the command register). A high to low pulse is given at EN pin when data is sent.

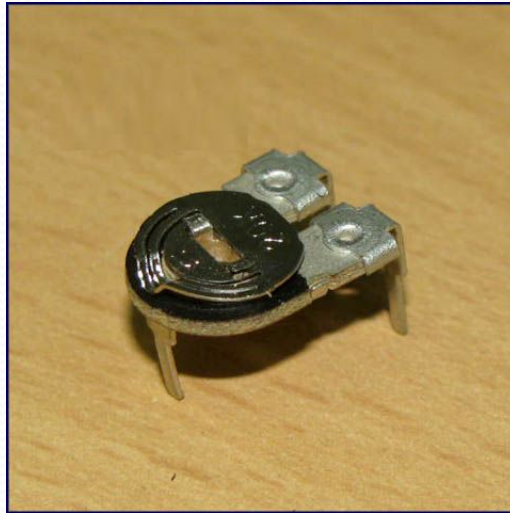
Conti...

- Displaying single character 'A' on LCD
- The LCD is interfaced with microcontroller (AT89C51). This microcontroller has 40 pins with four 8-bit ports (P0, P1, P2, and P3). Here P1 is used as output port which is connected to data pins of the LCD. The control pins (pin 4-6) are controlled by pins 2-4 of P0 port. Pin 3 is connected to a preset of 10k Ω to adjust the contrast on LCD screen. This program uses the above concepts of interfacing the LCD with controller by displaying the character 'A' on it.

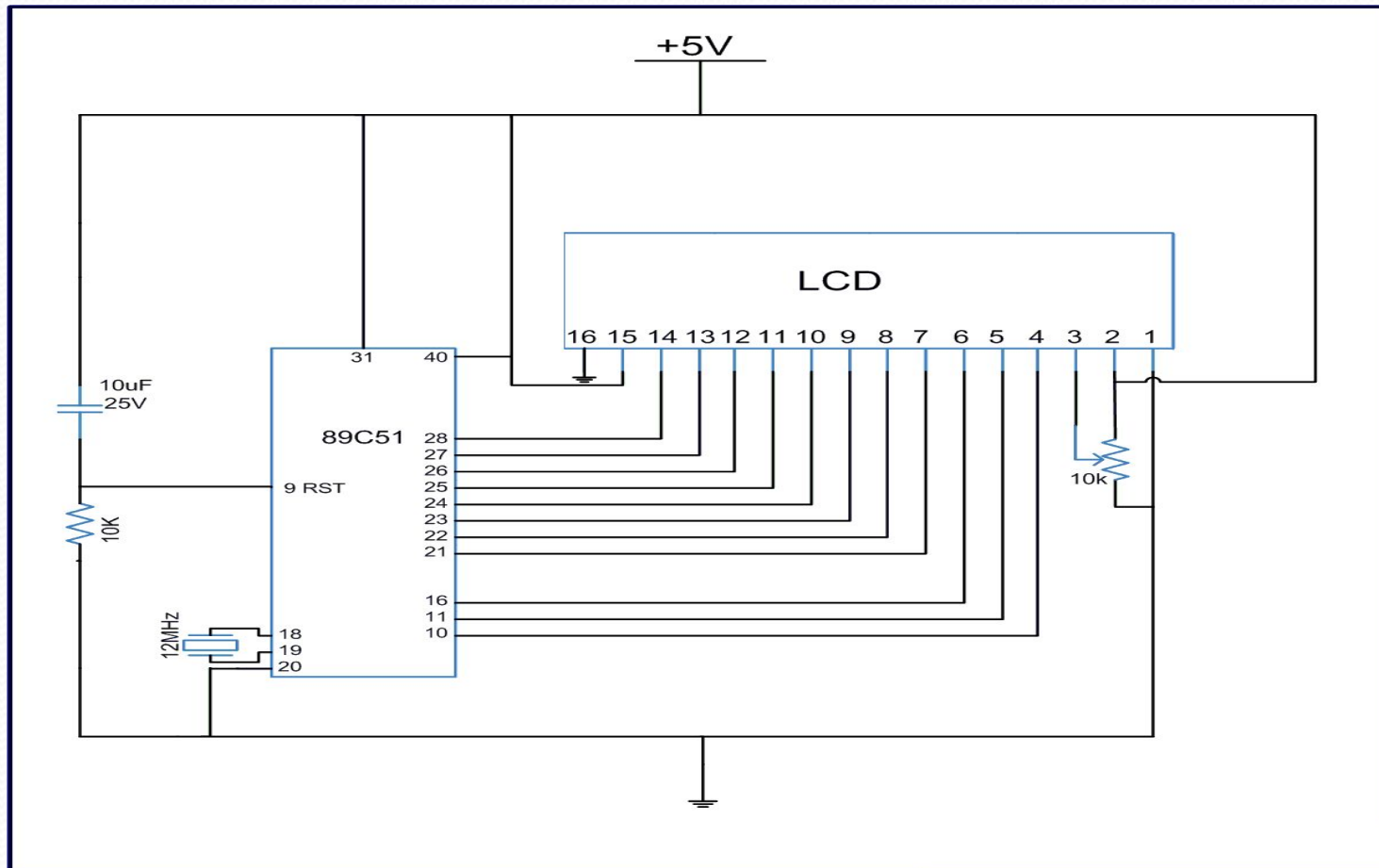
LCD



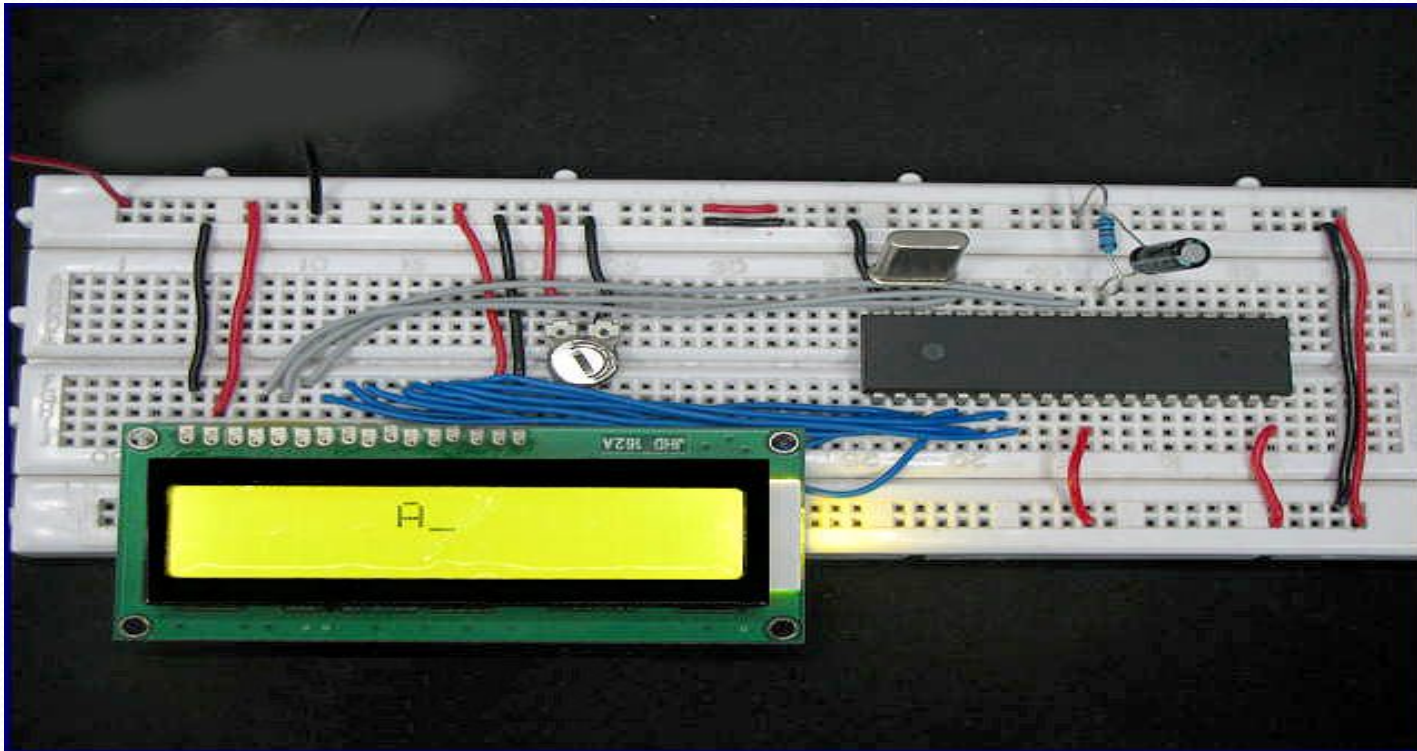
Preset



Circuit Diagram



Hardware



Coding

[LCD Code.docx](#)

Conti...

- [embedded unit 3.pdf](#)